

Question ID: ba8ca563

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A cube has a volume of **474,552** cubic units. What is the surface area, in square units, of the cube?

Correct Answer: 36504

Rationale

The correct answer is **36,504**. The volume of a cube can be found using the formula $V = s^3$, where s represents the edge length of a cube. It's given that this cube has a volume of **474,552** cubic units. Substituting **474,552** for V in $V = s^3$ yields $474,552 = s^3$. Taking the cube root of both sides of this equation yields $78 = s$. Thus, the edge length of the cube is **78** units. Since each face of a cube is a square, it follows that each face has an edge length of **78** units. The area of a square can be found using the formula $A = s^2$. Substituting **78** for s in this formula yields $A = 78^2$, or $A = 6,084$. Therefore, the area of one face of this cube is **6,084** square units. Since a cube has **6** faces, the surface area, in square units, of this cube is **6(6,084)**, or **36,504**.